

A BIOMECHANICAL STUDY

Speed, Take-off, and Handpush-off Abilities in Gymnastics Vaulting

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ABSTRACT

The present biomechanical study was conducted, on eight national level gymnasts, to find out the effect of speed, take-off and handpush-off ability, on selected biomechanical parameters, in vaulting event of Artistic Gymnastics. The basis of biomechanical analysis were Hand Spring 1½ Tucked Front Somersault and Tsukahara Vaults, respectively. The level of motor ability/physical pre requisites was measured using standard tests and the biomechanical parameters of the vaults, the horizontal approach run velocity, angular momentum after take off, and the rise of center of gravity of the gymnast in second flying phase, were obtained using established biomechanical investigational procedures. Study clearly indicated that the better gymnasts are better in speed, take-off and handpush-off abilities; are in a position to acquire higher value of horizontal approach run velocity, higher amount of angular momentum at take off, and they are also able to achieve higher rise of c.g., after leaving the horse. Thus, the entire vault performance is facilitated. It is, therefore, suggested to develop these abilities specially for vaulting.

KEY WORDS

Breast Stroke, Stroke Length, Stroke Frequency, Turn Time, Start Time

INTRODUCTION

Artistic Gymnastics is a popular discipline in the competitive world of sports. Vaulting Horse is one of its self-standing event.

According to general conception, motor abilities are the main and basic pre requisites for learning, execution of movement action, and for attaining the sport performance, respectively (Thiess & Schnabel, 1986). They can also be understood as 'complex part of prevalent physical pre requisites of man for achieving a sport performance through movements' (Marhold, 1987). These physical pre-requisites develop on the basis of anatomical, physiological, biomechanical, and psychological

peculiarities of the humans, in the process of their activities. They are the essential pre-requisites for overcoming the demands of training and competitions, as well as, their results (Group Authors, 1979). It means, besides sports technique and its mastery, sport motor abilities, which can be divided into two basic classes; the conditioning (energetic) motor abilities and co-ordinative (informational) motor abilities, play a very big role in laying out the foundation, and for developing special strength abilities, for the development of the performances, in Gymnastics (Hartig. & Buchmann, 1988; Fetzer, Knoll & Krug, 1982)

The trend of development in

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international vaulting is oriented on the realisation of such spectacular superior and difficult vaults which have additional rotations round transverse or/and longitudinal axis of the gymnast. The successful practical realisation of such vaults makes definite mechanical demands, (Hess, 1988). Kinetic energy, developed during approach run, is the most initial mechanical demand for the vault, the part of which is further transformed into rotational energy (angular momentum) during take-off phase of the vault, which is either further increased or decreased or maintained, as per the type of vault, during hand support phase of the vault. It may be, in general, understood that the motor ability and physical pre-requisites-speed, take-off and handpush-off abilities, respectively, of the gymnast, may influence mechanical demands of the vault. But, the available literature reveals that, so far, no study has been conducted in this direction. The present study is taken with the hypothesis that the speed take-off as well as the handpush-off ability are the important determinant factors for a better performance in vaulting, as they help in acquiring the required, mechanical demands, as per the vaults; and the level of these abilities draw a distinct differentiating line in between the better and poor gymnasts.

METHODOLOGY

Handspring 114 Tucked Front Somersault and Tsukahara Vaults performed by 8 national level gymnasts, attending the national camp, were selected for biomechanical analysis. Biomechanical films of these vaults were prepared during the national camp at NIS Patiala, with a movie camera of type Biomechanics 500, working at 100 frames per second, and

placed perpendicular to the plane of the vault movement. The performance of each gymnast was also judged simultaneously, by experienced national judges. On the basis of this evaluation, the gymnasts were divided into two groups: first best four gymnasts G_1 and rest four gymnasts G_2 on Handspring 114 Front Somersault and Tsukahara Vaults, respectively. The prepared films were analysed biomechanically according to the established steps (Knoll & Schaffer, 1984; Marhold, 1986).

The speed ability of the gymnast was measured cinematographically by determining maximum running velocity (V) on 25 meter distance. The standing vertical jump was chosen to measure the take-off ability of the gymnast. The vertical height, (H) reached in jump, was the expression for this ability. The level of handpush-off ability of the gymnast was determined with the help of hand-push-off movement executed on gymnastics rolling bench (Sharma, 1990). The maximum distance covered (S) by the gymnast was considered as the indicator of handpush-off ability of the gymnast.

To determine the correlation between the motor ability/ physical prerequisite and sport performance selected biomechanical parameters, the statistical procedure rank was used & (Clauss & Ebner, 1983).

RESULTS & DISCUSSION

Table 1 shows the average values of test results of the corresponding motor ability and physical pre-requisites of both the groups of Indian gymnasts, together with the values of biomechanical parameters: the horizontal velocity before reaching the hoise, angular momentum at take-off, and rise of center of gravity of the gymnast after leaving the horse.

Table-1: Average values of speed, take-off and handpush off abilities and biomechanical parameters of the Vaults : horizontal approach run velocity (v) angular momentum at take off (L) and rise of c.g. of gymnast after leaving horse (h), of first four ranking (G1, G2) gymnasts.

Group	Speed Ability V(m/s)	Take-off Ability H (m)	Handpush-off Ability S (m)	V (m/s)	(Kgm ² /s)	h(m)
Handspring 1% Salto Vault						
G ₁	8.67	0.63	0.49	7.65	81.0	0.40
G ₂	8.46	0.57	0.42	7.44	78.0	0.31
Tsukahara Vault						
G ₁	8.61	0.61	0.48	7.59	68.5	0.19
G ₂	8.37	0.60	0.43	7.37	62.5	0.14
Measurement error	±0.15	±0.02	±0.01	±0.12	±4.5	±0.02

From Table 1 it is clear that first four ranking gymnasts, in case of both vaults, have higher average values of test results in comparison to the last four ranking gymnasts. This result gives the evidence that the gymnasts of higher performance level in vaulting possess a higher level of conditioning, motor ability, and corresponding physical pre-requisites.

Horizontal approach run velocity prior to the Reuther Board contact, angular momentum as a result of take-off, and the rise of c.g. of gymnast after leaving the Horse contact, are some biomechanical parameters which ultimately facilitate the second flying phase of the Vault, (Knoll, 1980). They are the indirect expressions of the basis of the conditioning motor abilities. From the comparison of the values in Table

1, it is further evident, in general, that the gymnasts with higher average value of horizontal approach run velocity, angular momentum after take-off, and the rise of c.g. after leaving the Horse in second flying phase, possess a higher level of speed, take-off and handpush-off ability. From the analysis, it is also clear that the lower values of mentioned biomechanical parameters lie in the insufficient level of gymnast's motor ability and physical pre-requisites.

To confirm the above results more precisely, the rank correlation coefficients were determined between the results of various abilities and the level of Vault performances and various biomechanical parameters (see Table II).

Table-2: Correlation coefficients (r) between the results of motor ability, physical pre-requisites and the value of selected biomechanical parameters of the vaults.

Correlation between	Handspring 1½ Salto Vault		Tsukahara Vault	
	r	Significance level	r	Significance level
Speed ability and horizontal approach run velocity	0.934	1%	0.833	1%
Take-off ability and angular momentum at take-off	0.810	5%	-	-
Handpush-off ability and rise of c.g. on leaving horse	0.934	1%	0.667	5%
Combine motor ability test and combined biomechanical parameters	0.858	1%	0.681	5%

Table 2 reveals that in case of Handspring Salto Vault, a very high correlation (1 percent level of significance) exists between the speed ability of the gymnast and his achieved horizontal approach run velocity, during the performance of the Vault. Exactly, the same highest correlation (1 per cent level of confidence) is found between the results of handpush-off ability of the gymnast and his achieved height of rise of c.g. after leaving the horse, as well as, between the results of combined level of his motor ability and physical pre-requisites and combined result of biomechanical parameters: height of the flight and maximum flight distance of second flight phase. A correlation (5 per cent level of significance) exists between the take-off ability and angular momentum at take-off. Almost similar correlation is found in case of Tsukahara Vault (see Table 2)

The above statistical procedure, rank correlation underlines the importance of speed, take-off and handpush-off abilities for higher values of biomechanical parameters of the vaults: horizontal approach run velocity reaching on the

Reuther Board, angular momentum at take-off and the rise of c.g. of the gymnast after leaving the horse.

CONCLUSION

The comparison of the results of the test of speed, take off and handpush off abilities of Indian best and poor gymnasts indicates that the better gymnasts in vaulting performance are better in their level of speed, take-off, and handpush-off abilities, in comparison of poor gymnasts. This study also confirms the fact that the Indian gymnasts in vaulting, which are better in speed, take off, and handpush-off abilities in comparison of poor vaulters, are in a position to acquire better initial mechanical demand of a vault. They are also in a position and capable of transforming these demands further more efficiently and adequately by utilizing their take-off abilities. Consequently, it is suggested and recommended that to build high class vaulter, in Gymnastics, the speed, the take-off and handpush-off abilities must get a proper scientific approach for their development during the training process.

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